Gaussian Random Variable

L8_S2_A.pdf
PDF

Purpose

Prepare a formal lecture scribe using the provided lecture slides. The scribe must serve as exam reference material and contain definitions, notations, assumptions, preliminaries, main results, proofs, conclusions, and worked examples exactly as presented in the lecture slides. Use only the lecture content. Do not add explanations, interpretations, assumptions, or examples not present in the lecture slides.

Example Lecture Scribe (Structure Only)

Topic: Fundamental Properties of Sequences

1) Definitions and Notations
Definition 1.1 (Sequence)

A sequence is an ordered list of elements indexed by natural numbers.

Notation

Let $\{a_n\}$, $n = 1,2,3,\ldots$

where

a_n = nth element of the sequence

2) Assumptions and Preliminaries

The sequence is defined for all natural numbers.

Each index corresponds to exactly one element.

3) Main Result
Proof
a_n = c

where

c = constant

Then

a_1 = c, a_2 = c, ...

Thus all elements are equal.

Conclusion

A constant sequence has identical values for all indices.

4) Working Examples
Example 1
a_n = 5

Then

a_1 = 5, a_2 = 5
Instruction

Using the exact same structure as the example above, generate a lecture scribe from the provided lecture slides.

Strict Requirements

You MUST follow ALL of these rules:

Use EXACTLY the same structure and section order:

Definitions and Notations

Assumptions and Preliminaries

Main Result

Proof

Conclusion

Working Examples

Use ONLY the content present in the lecture slides.

Do NOT add:

External textbook content

Extra explanations

Additional examples

Interpretations

Assumptions not explicitly stated

If any section is not present in the lecture slides, leave that section blank.

Preserve all mathematical expressions, formulas, definitions, derivations, and examples exactly as presented.

Maintain formal academic tone suitable for exam reference material.

Do NOT include commentary, reasoning, or explanatory text outside the lecture content.

Lecture Scribe – Lecture 8 (CSE400)
Topic: Gaussian Random Variable, Standard Forms, and Relation between Φ-function and Q-function

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1) Definitions and Notations

Definition 1.1 (Gaussian Random Variable)

A Gaussian random variable is one whose PDF can be written in the general form:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

Notation

Mean:

$$m$$

Variance:

$$\sigma^2$$

Standard deviation:

$$\sigma$$

CDF:

$$F_X(x)$$

Gaussian CDF relation:

$$F_X(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-m)^2}{2\sigma^2}\right) dy$$

Standard normal CDF (Φ-function):

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} \exp\left(-\frac{t^2}{2}\right) dt$$

Gaussian tail function (Q-function):

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_{x}^{\infty} \exp\left(-\frac{t^2}{2}\right) dt$$

Error function:

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_{0}^{x} \exp(-t^2)\, dt$$

Complementary error function:

$$\operatorname{erfc}(x) = 1 - \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_{x}^{\infty} \exp(-t^2)\, dt$$

Key identity:

$$Q(x) = 1 - \Phi(x)$$

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2) Assumptions and Preliminaries

CDF evaluation:

$$F_X(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-m)^2}{2\sigma^2}\right) dy$$

Standardization:

$$= \int_{-\infty}^{\frac{x-m}{\sigma}} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt$$

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3) Main Result

Gaussian CDF relation:

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

Tail probability relation:

$$Pr(X > x) = \int_{\frac{x-m}{\sigma}}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt$$

$$= Q\left(\frac{x-m}{\sigma}\right)$$

Key identities:

$$Q(x) = 1 - \Phi(x)$$

$$F_X(x) = 1 - Q\left(\frac{x-m}{\sigma}\right)$$

Proof

Evaluating the CDF:

$$F_X(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-m)^2}{2\sigma^2}\right) dy$$

Substitute:

$$t = \frac{y-m}{\sigma}$$

$$= \int_{-\infty}^{\frac{x-m}{\sigma}} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt$$

$$= \Phi\left(\frac{x-m}{\sigma}\right)$$

Tail probability:

$$Pr(X > x) = \int_{\frac{x-m}{\sigma}}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt$$

$$= Q\left(\frac{x-m}{\sigma}\right)$$

Conclusion

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

$$Pr(X > x) = Q\left(\frac{x-m}{\sigma}\right)$$

$$Q(x) = 1 - \Phi(x)$$

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4) Working Examples

Example 1

A random variable X has PDF:

$$f_X(x) = \frac{1}{\sqrt{8\pi}} \exp\left(-\frac{(x+3)^2}{8}\right)$$

Comparing with Gaussian PDF:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

we identify:

$$m = -3$$

$$\sigma^2 = 4$$

$$\sigma = 2$$

Gaussian relations:

$$F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$$

$$Pr(X > x) = Q\left(\frac{x-m}{\sigma}\right)$$

(a) Find $Pr(X \leq 0)$

$$Pr(X \leq 0) = F_X(0)$$
$$= \Phi\left(\frac{0-m}{\sigma}\right)$$
$$= \Phi\left(\frac{3}{2}\right)$$
$$= 1 - Q\left(\frac{3}{2}\right)$$

(b) Find $Pr(X > 4)$

$$Pr(X > 4) = Q\left(\frac{4-m}{\sigma}\right)$$
$$= Q\left(\frac{7}{2}\right)$$

End of Lecture Scribe